

Hereditary butyrylcholinesterase deficiency

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Pipeline complete — awaiting your sign-off

Repurposing hypothesis: PHYSOSTIGMINE → Hereditary butyrylcholinesterase deficiency

Repurposing-only pool: the candidate compounds for this target were restricted to FDA-approved / known drugs (ChEMBL max_phase $\geq$ 4) at collection time. Unapproved research-grade tool compounds were excluded from the pool, not merely down-ranked.

1. Hypothesis summary

PHYSOSTIGMINE is proposed as a repurposing candidate against **Hereditary butyrylcholinesterase deficiency** via the target **BCHE**. It shows a ChEMBL median pChEMBL affinity of 7.46 at assay confidence 9/9, an Open Targets target-disease association of 0.805, and a Tanimoto similarity of 0.786 to PHYSOSTIGMINE SALICYLATE. Target network context (BioGRID, physical/genetic — not mechanism): BCHE, COLQ, PTPRK, Sgol2, LOXL1, ENTPD5, LTBP1, NAF1. The resulting composite score is 0.8201.

Chemist rationale: Physostigmine has a median pChEMBL affinity of 7.46 against BCHE, measured at assay confidence score 9, and is an approved or known drug with a Tanimoto similarity of 0.786 to the

nearest approved drug physostigmine salicylate. The target BCHE has eight recorded BioGRID physical or genetic interactors: BCHE, COLQ, PTPRK, Sgol2, LOXL1, ENTPD5, LTBP1, and NAF1.

STAGE 1 PRIORITIZATION SCORES

- **tractability_score:** 0.5854 (ChEMBL bioactivity + AlphaFold pLDDT – prior-trial-failure penalty)
- **unmet_need_score:** 1.0000 (treatment availability + prevalence)

These are computed by the same formulas used to rank the full rare-disease / NTD universe; a manually chosen target is scored identically, never faked or skipped.

2. Evidence table

Evidence	Value
ChEMBL median pChEMBL affinity	7.46
Assay confidence score (0-9)	9
Open Targets association score	0.805
Tanimoto to nearest approved drug	0.786 (PHYSOSTIGMINE SALICYLATE)
Approved / known drug	yes
Mutation-specific approved indication (disclosure)	No specific mutation named in approved indication
Lipinski/Veber (MW, logP, HBD, HBA, TPSA, rotB)	275.4, 1.77, 1, 4, 44.8, 1
Lipinski violations / Veber pass	0 / yes

Evidence	Value
AFDB apo structure mean pLDDT (free protein, no ligand)	93.4
Boltz structure confidence (0-1)	0.871
Boltz binding-pose confidence (0-1)	0.450
Boltz predicted affinity (relative optimization score, 0-1, NOT a Kd)	0.153
Boltz predicted structure (CIF)	Download CIF (/api/structures/d09c3ac47b03f187fcdc063f07eee6cd.cif)
Boltz ADME – lipophilicity (logD)	0.817
Boltz ADME – permeability	1.833
Boltz ADME – solubility	high-confidence
openFDA adverse-event signal (FAERS)	DELIRIUM (30), ANTICHOLINERGIC SYNDROME (29), DRUG INTERACTION (20), OVERDOSE (20), TOXICITY TO VARIOUS AGENTS (17)
Prior trials for this exact drug+disease	0
Target discovery method	genetic_association

3. Full source citations

- **PMIDs (6):** 18075469, 29631548, 30593143, 38420074, 39814397, 40778538
- **ChEMBL activity IDs (23):** 117581, 1191527, 1206877, 1284248, 1413695, 1491404, 1699449, 1808320, 1808321, 1840539, 226310, 2367616, 2435546, 350518, 350519, 361652, 361653, 454432, 486565, 683108, 728066, 731682, 772462

- **NCT numbers (0):** none

4. Composite score breakdown

Term	Weight	Component value	Contribution
Normalized pChEMBL affinity	0.30	0.637	0.1911
Assay confidence (score / 9)	0.20	1.000	0.2000
Normalized Open Targets association	0.20	0.805	0.1611
Normalized Tanimoto similarity	0.15	0.786	0.1179
No prior failed trial (1/0)	0.15	1	0.1500
Composite (weighted sum - penalty - cap)			0.8201

Weighted sum before penalty = 0.8201; penalty = 0.0000; reported composite_score = 0.8201.

5. Limitations

- **Binding is not efficacy.** A high binding-pose confidence (0.450) or predicted affinity (0.153) only suggests the molecule may occupy the target; it does NOT establish agonism vs. antagonism, functional modulation, or therapeutic benefit.
- **ADME values are model predictions, not measurements.** The Boltz lipophilicity/permeability/solubility numbers are computed estimates and must be confirmed experimentally before any decision.
- **Structure confidence is bounded.** This hypothesis relies on a Boltz structure_confidence of 0.871 (complex pLDDT 0.913) and an AFDB apo mean pLDDT of 93.4; the AFDB model contains NO ligand, so the protein-ligand pose is entirely a Boltz prediction.

- **Assay-type and species caveats.** The affinity is a median pChEMBL over Homo sapiens IC50/Ki assays at confidence ≥ 8 ; assay heterogeneity and the bounded approved-drug reference set for Tanimoto still apply.
- **Absence of evidence is not evidence of absence.** A zero prior-trial count or no adverse-event signal may reflect that the pair has simply never been studied, not that it is safe or untried.
- **This is a repurposing *hypothesis*, not a finding.** It is a prioritised starting point that requires wet-lab and, ultimately, clinical validation.

TARGET DRUGGABILITY CONTEXT

- **Approved drugs with known mechanism against this target (ChEMBL):** 4 — PROPANIDID, RIVASTIGMINE, RIVASTIGMINE TARTRATE, TACRINE HYDROCHLORIDE
- **Historical difficulty literature:** insufficient signal found (fewer than 2 qualifying abstracts in targeted PubMed searches for undruggability / resistance / difficulty).

Druggability context is informational only. It does not affect tractability_score, unmet_need_score, composite_score, or STRONG_MATCH.

HUMAN REVIEW CHECKPOINT

This is a falsifiable hypothesis, not a finding.

Record your scientific reasoning below, then sign off. Approving advances this candidate for wet-lab validation; rejecting closes the case permanently.

REVIEWER NOTE **required**

e.g. Affinity and structure confidence justify wet-lab follow-up despite the sub-threshold composite score.



Approve case

Reject case